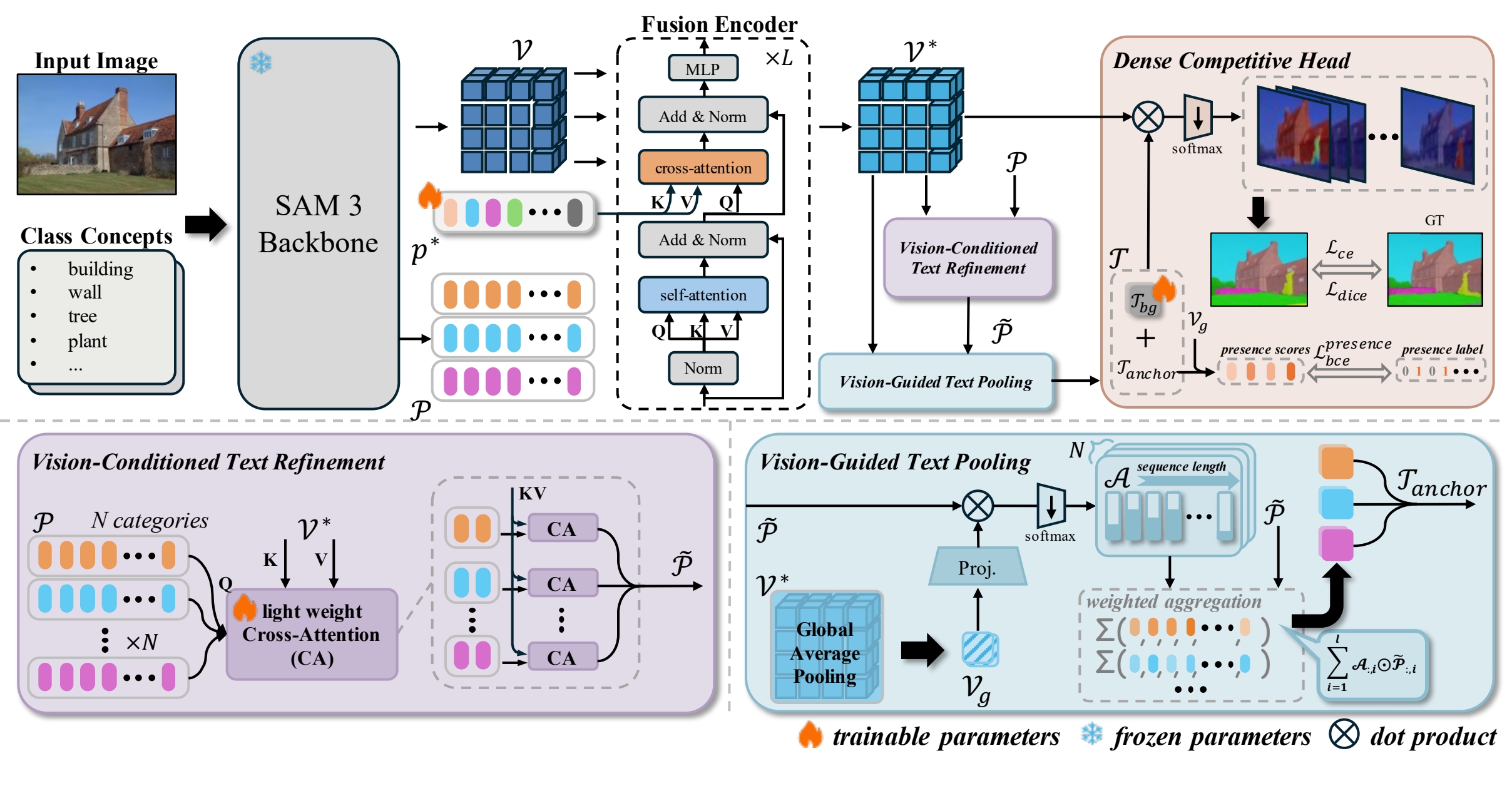

Input Image
Class Concepts
building
wall
tree
plant
...
SAM 3 Backbone
Fusion Encoder
MLP
Add & Norm
cross-attention
Add & Norm
self-attention
Norm
×L
K V Q
Q K V
Vision-Conditioned Text Refinement
Vision-Guided Text Pooling
Dense Competitive Head
softmax
GT
presence scores
presence label
Vision-Conditioned Text Refinement
N categories
light weight Cross-Attention (CA)
CA
CA
CA
KV
Vision-Guided Text Pooling
sequence length
Proj.
softmax
Global Average Pooling
weighted aggregation
trainable parameters
frozen parameters
dot product